

Mystery Meat Analysis

BIOLM0051

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Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

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Introduction

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Methods

Pipeline Overview

Some python code:

Results

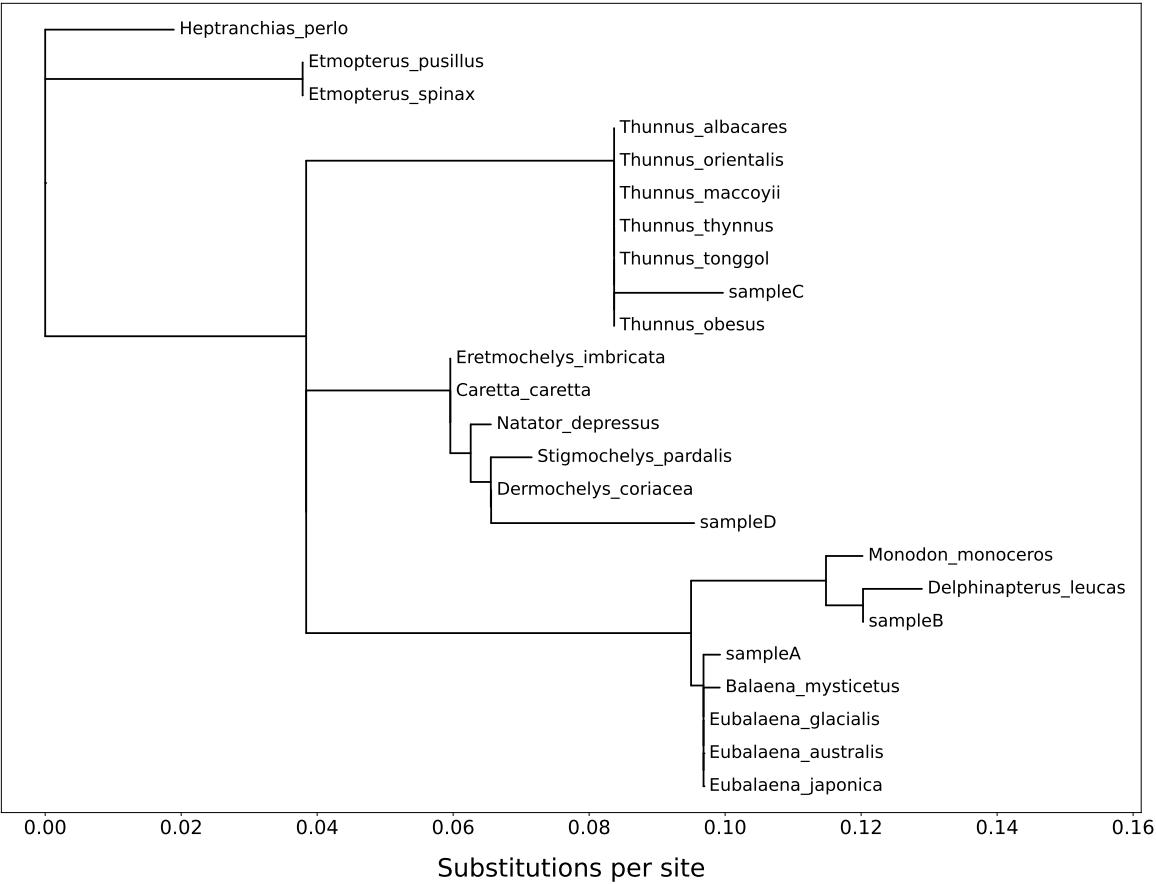


Figure 1: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and their top BLAST hits

Discussion

sampleD_part1 nt length is 524bp, whereas the efetch FASTA lengths are 525bp. This single deletion could be causing the frame shift visible towards the end of the GAINFITTIIN region in Fig. 2. This deletion may have been caused by an error during sequencing.

Figure X. Maximum-likelihood tree from the original protein alignment. Figure Y. Maximum-likelihood tree from a manually corrected alignment in which one ambiguous base - N - was inserted into sampleD_part1 at the suspected deletion site to restore the reading frame. This tree is shown to assess the impact of the putative sequencing error on branch length and topology.

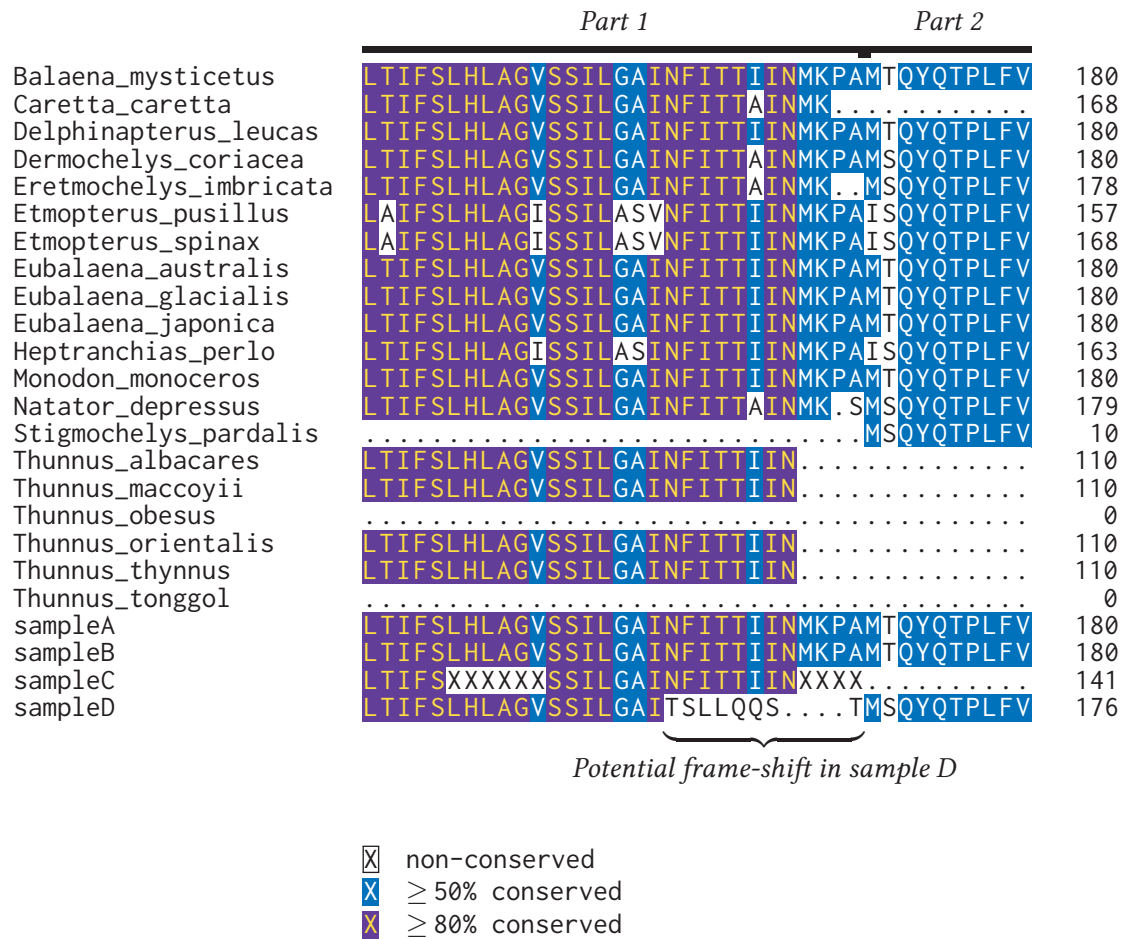


Figure 2: Subset of supermatrix alignment with potential frame shift error in sampleD near the highly-conserved INFITTIIN region.



Figure 3: Nucleotide Needleman-Wunsch pairwise global alignment of the tail end of sampleD_part1 against closest reference, *Dermochelys coriacea*, and showing a possible sequencing base miss at position 486 in sampleD_part1.

References

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Figures for Manual Correction of Suspected Sequencing Error

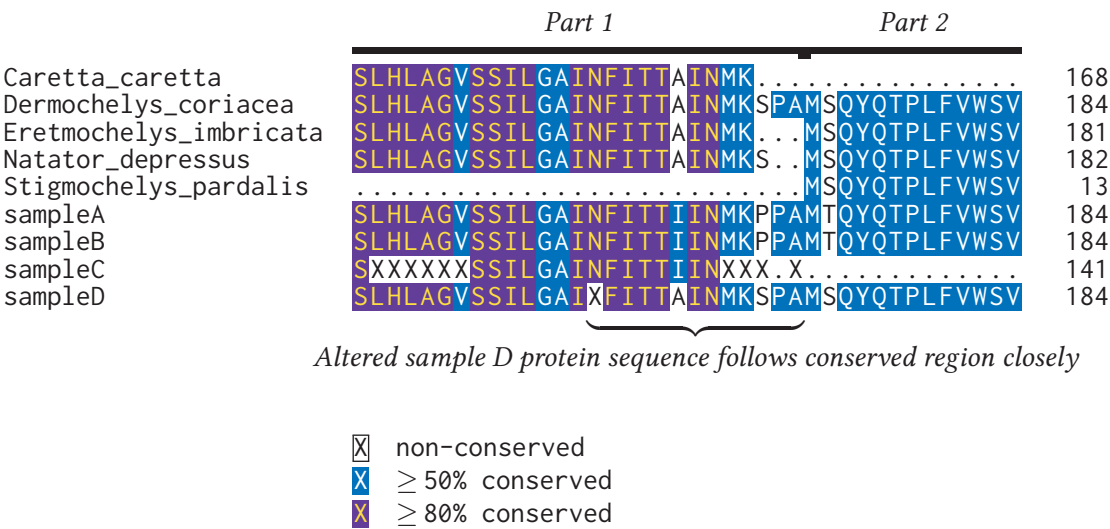


Figure 4: Subset supermatrix alignment after manually correcting sampleD_part1 with one ambiguous base (N) inserted in the nucleotide sequence (other references left out of figure still contribute to the conserved residue %). The altered protein sequence from one ambiguous base addition in the nucleotide sequence aligns more closely to highly-conserved INFITTIIN region seen across other sequences compared to Figure 2.

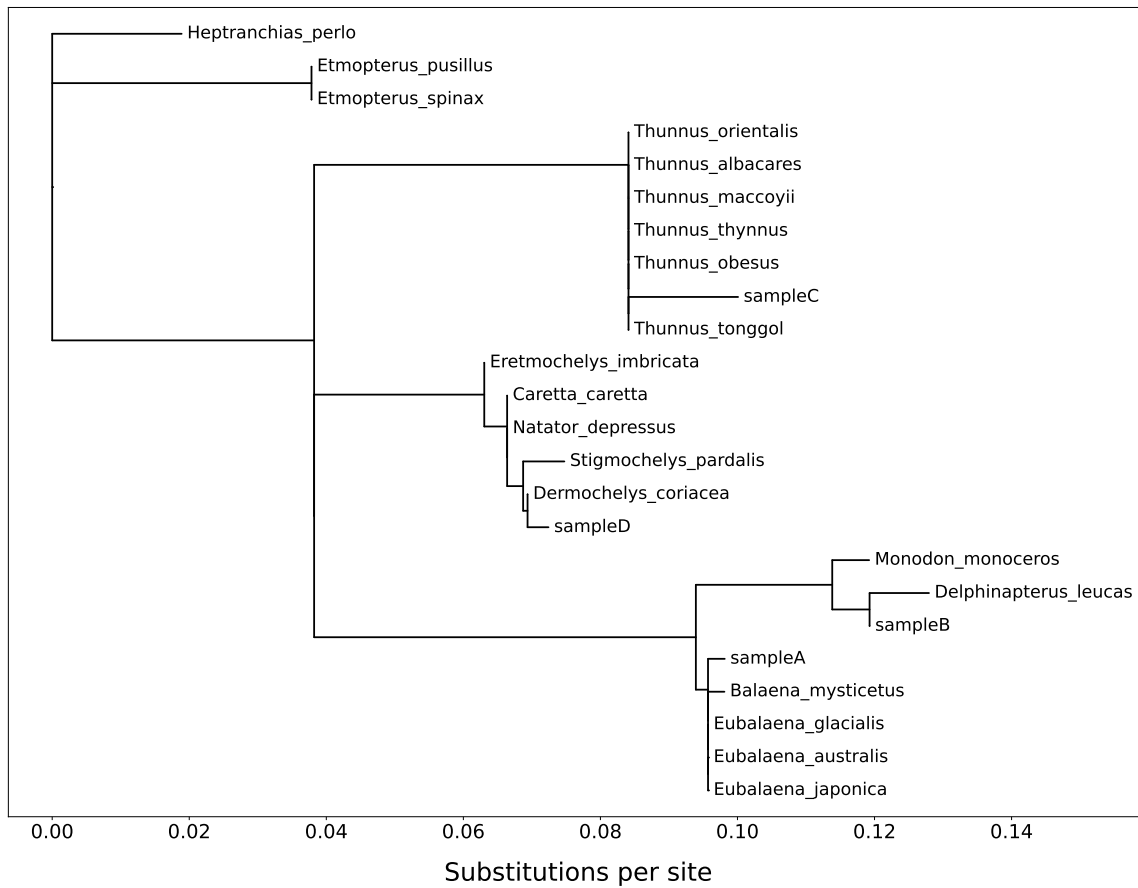


Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into sampleD-part1. The resulting branch length of sampleD is much shorter than in Figure 2, indicating that a frame shift due to a sequencing base miss caused the extended branch length from the rest of the turtle clade.